

Rajarata University of Sri Lanka

Department of Computing

ICT2403 – Graphics and Image Processing

Submission 03 – Frame Enhancement Method

1. Introduction to Frame Enhancement

Road damage detection from video footage is a challenging computer vision task that heavily relies on the quality of input frames. Raw video frames captured in real-world conditions often suffer from a range of degradations – including noise, poor illumination, and motion blur – that can severely hinder the performance of downstream segmentation and detection algorithms.

Image enhancement is therefore a critical pre-processing stage in any road surface analysis pipeline. By systematically improving frame quality before analysis, enhancement techniques allow the system to detect subtle features such as cracks, potholes, and surface deformations with significantly greater accuracy.

Importance of Enhancement in Road Damage Detection

- Road damage features such as fine cracks and hairline fractures occupy only a small pixel area and can be easily masked by noise or low contrast.
- Enhancement increases the signal-to-noise ratio of the image, making the target region of interest (ROI) more distinguishable from the background.
- Accurate segmentation algorithms (e.g., thresholding, edge detection, deep learning models) are highly sensitive to input image quality; enhanced frames yield fewer false positives and missed detections.

Challenges in Raw Video Frames

- Sensor noise introduced by the mobile camera sensor, especially in low-light conditions.
- Compression artefacts introduced during MP4 encoding that degrade high-frequency edge information.
- Shadows from trees, vehicles, and overhead structures causing uneven illumination across the road surface.
- Motion blur from handheld recording or vehicle vibrations reducing sharpness of captured frames.

Relationship Between Enhancement and Accurate Segmentation

A well-designed enhancement pipeline produces frames with uniform illumination, reduced noise, and preserved edge detail. These properties directly contribute to improved segmentation accuracy by ensuring that edge-based and region-based segmentation algorithms can reliably distinguish road damage from intact pavement.

2. Identification of Frame Degradations

A systematic analysis was conducted on the extracted video frames to identify and quantify the types of degradations present. Each frame was converted to grayscale and evaluated using the `analyse_frame()` function defined in the pipeline, which computes four key quality metrics:

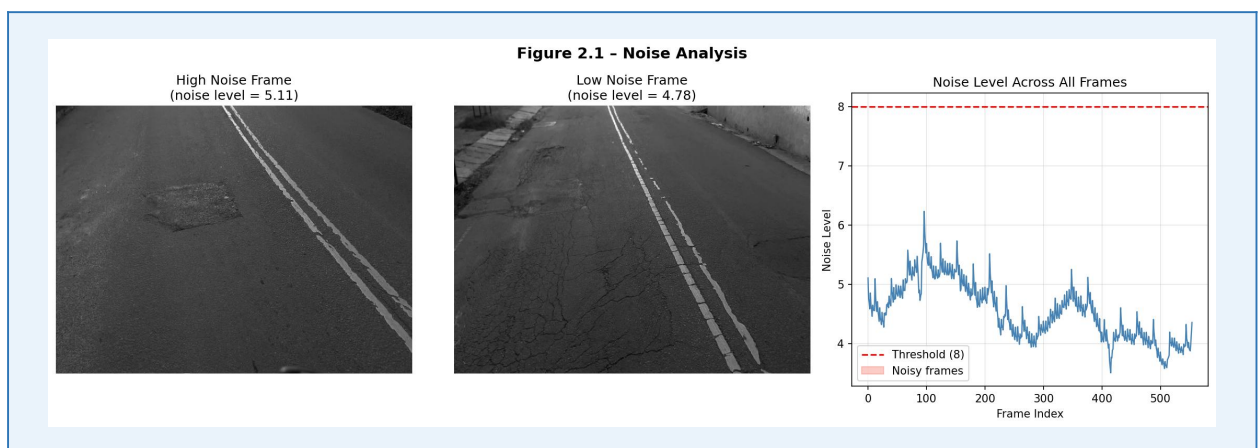
- Mean Brightness – average pixel intensity (range 0–255); values below 70 indicate underexposure, above 190 indicate overexposure.
- Standard Deviation of Pixel Intensities (Contrast) – values below 45 indicate insufficient contrast.
- Laplacian Variance (Sharpness) – values below 80 indicate blurred or insufficiently sharp frames.
- Noise Level – estimated by computing the standard deviation of the residual between the original and a Gaussian-blurred version; values above 8 indicate significant noise.

2.1 Noise Analysis

Noise in the captured road damage video arises from three primary sources:

- Sensor noise: Random pixel-level intensity variation inherent to the mobile camera sensor, particularly visible in low-texture regions such as flat road surfaces.
- Compression noise: Block artefacts and high-frequency distortions introduced during MP4 video compression (MPEG codec) that can be mistaken for surface damage features.
- Low-light noise: Increased random grain in frames captured under poor illumination, where the sensor applies higher ISO amplification.

The noise level for each frame was estimated using the residual between the raw frame and its Gaussian-blurred version (kernel size 3×3 , $\sigma=0$). A noise level exceeding 8 threshold units triggered the application of the Median Filter in the enhancement pipeline.



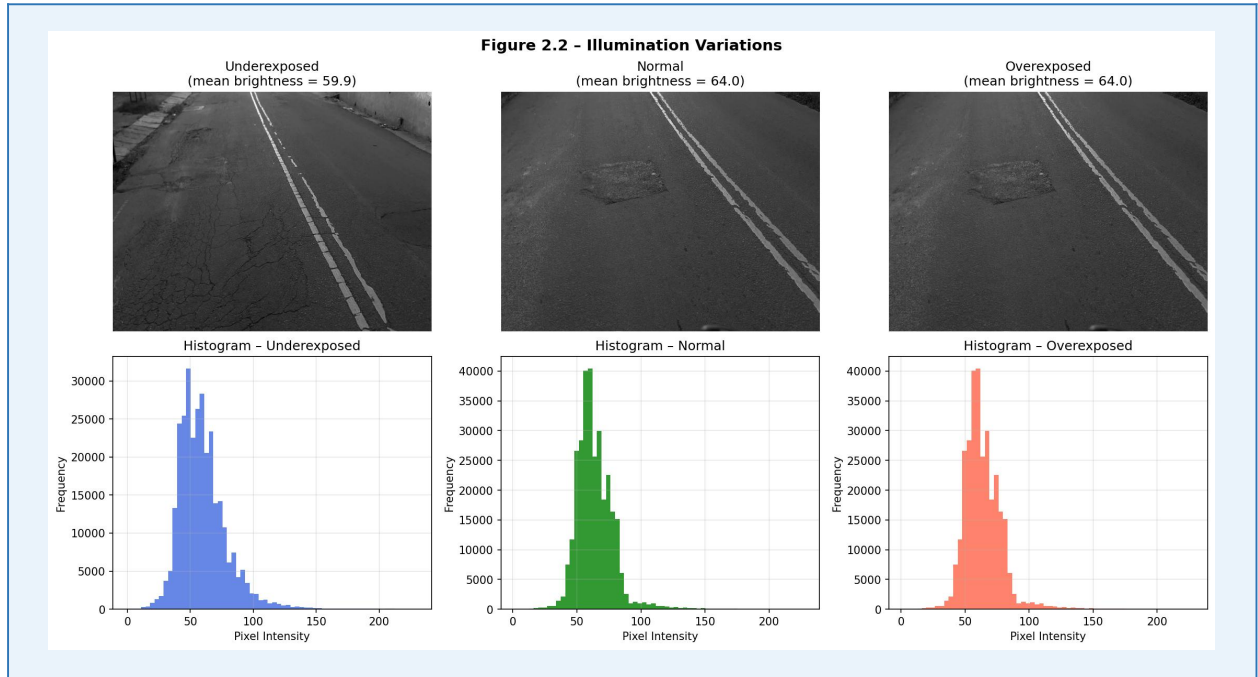
2.2 Illumination Variations

Illumination inconsistency is one of the most prevalent degradations observed in the road damage video. The following conditions were identified:

- Uneven lighting: Varying ambient light across different sections of the road caused non-uniform brightness gradients within individual frames.

- Shadows from trees and vehicles: Cast shadows create dark regions that can be misidentified as surface damage by segmentation algorithms.
- Overexposed regions: Areas with direct sunlight reflection on the road surface produce saturated (bright) patches that suppress detail.
- Underexposed regions: Low-light frames captured in shaded sections exhibit poor contrast and reduced feature visibility.

Mean brightness and contrast metrics were used to detect illumination issues. Frames with mean brightness below 70 or above 190 were flagged for Brightness Correction, while frames with a contrast standard deviation below 45 were enhanced using CLAHE.

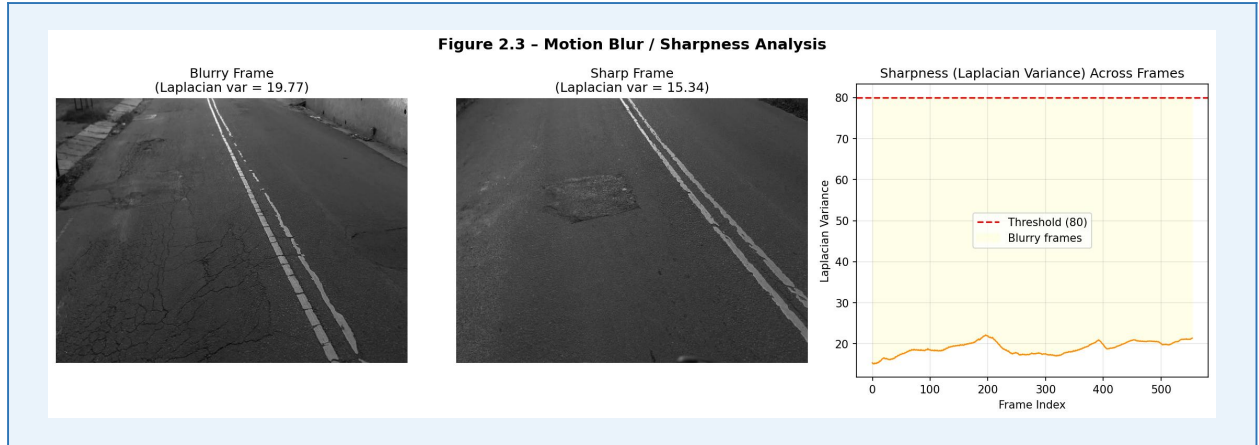


2.3 Motion Blur and Camera Instability

Motion blur arises from two sources in this project:

- Handheld recording effects: Small involuntary hand movements during capture introduce directional blur in individual frames.
- Moving vehicle vibrations: Vibrations from the recording platform cause repeated, low-amplitude frame displacements resulting in generalised blur.

Sharpness was quantified using the Laplacian variance operator (cv2.Laplacian with CV_64F precision). A Laplacian variance below 80 was used as the threshold to classify a frame as blurred and trigger the sharpening stage of the pipeline.



3. Pre-Processing of Video Frames

Before enhancement can be applied, the raw video file must be converted into individual processable image frames. This pre-processing stage was implemented using the OpenCV VideoCapture interface in Python.

3.1 Frame Extraction Method

The video file `Road_Damage.mp4` was read frame-by-frame using `cv2.VideoCapture()`. A while-loop iterated over all frames, reading each with `cap.read()` and checking the return flag `ret` to detect the end of the video stream.

A `frame_skip` parameter (set to 1 in the current implementation) controls the temporal sampling rate. Setting `frame_skip=1` extracts every frame; increasing this value would extract every *n*th frame, reducing the dataset size while maintaining temporal coverage.

3.2 Frame Format and Resizing

Each extracted frame was resized to a standardised resolution of 640×480 pixels using `cv2.resize()`. This ensures consistent spatial dimensions across all frames, which is a prerequisite for both the adaptive enhancement pipeline and any downstream segmentation algorithm. Frames were saved in JPEG format (.jpg) with default compression quality settings.

3.3 Frame Selection Strategy

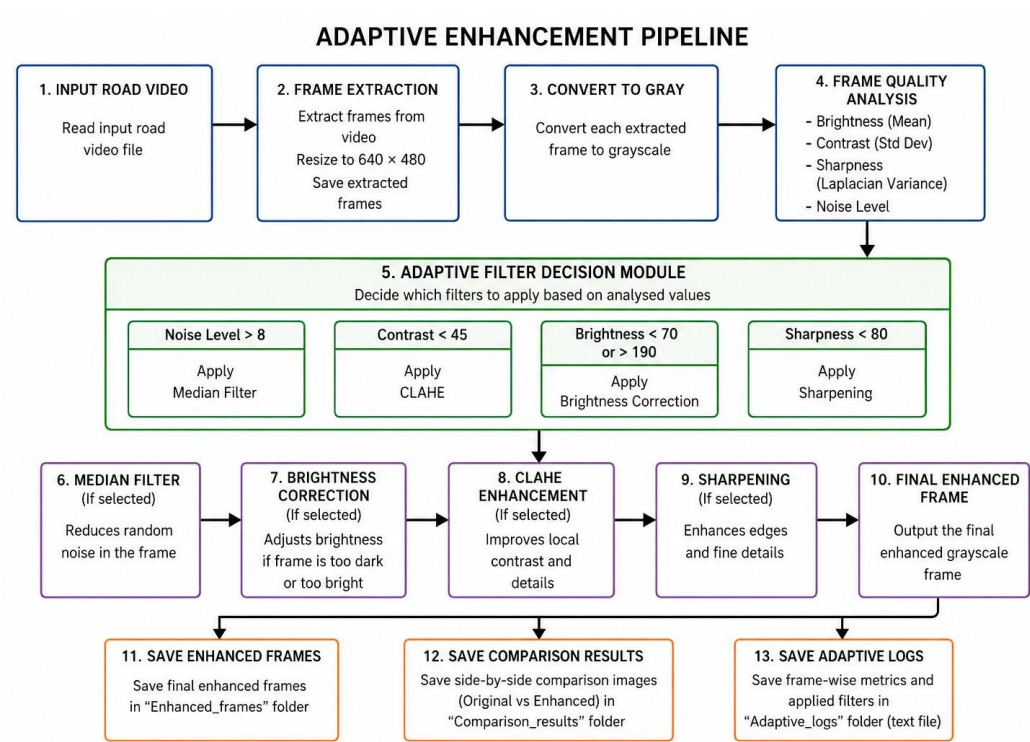
The current implementation extracts every frame (`frame_skip=1`), resulting in the full temporal dataset being available for enhancement. This strategy ensures no road damage features are missed due to temporal subsampling, at the cost of a larger dataset. The `frame_skip` parameter can be adjusted for efficiency in future deployments.



4. Proposed Image Enhancement Pipeline

The proposed adaptive image enhancement pipeline is the core contribution of this submission. Unlike a static pipeline that applies fixed filters to all frames, the proposed system first analyses each frame's quality metrics and then selectively applies only the filters that address the specific degradations detected. This adaptive strategy avoids over-processing well-captured frames while correcting degraded ones.

4.1 Enhancement Pipeline Flowchart



4.1 Grayscale Conversion

The first operation applied to each BGR colour frame is conversion to grayscale using `cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)`. Road damage features such as cracks and potholes are primarily characterised by intensity gradients rather than colour information. Grayscale processing reduces computational complexity and eliminates the influence of colour variations (e.g., road surface paint, wet/dry differences) that could interfere with damage detection.

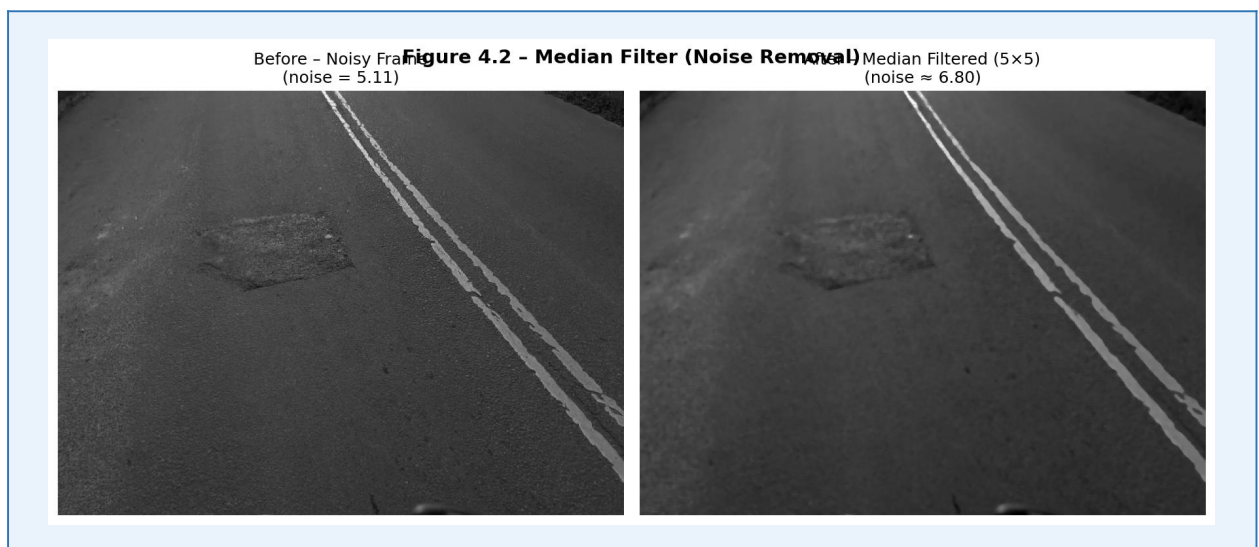
4.2 Stage 1: Median Filter (Noise Removal)

Trigger condition: `noise_level > 8`

The Median Filter is a non-linear spatial filter that replaces each pixel's intensity with the median value within a defined neighbourhood window. Unlike Gaussian or mean filters, the Median Filter is effective at removing impulse (salt-and-pepper) noise and compression artefacts without blurring edges – a critical property for preserving crack boundaries.

Implementation: `cv2.medianBlur(enhanced, 5)` – a 5×5 pixel neighbourhood is used, providing effective noise suppression while maintaining computational efficiency.

Mathematical operation: For a pixel at position (i,j), the output $I'(i,j) = \text{median}\{ I(i+s, j+t) : s,t \in [-2,2] \}$



4.3 Stage 2: Brightness Correction

Trigger conditions: `mean_brightness < 70` (underexposed) OR `mean_brightness > 190` (overexposed)

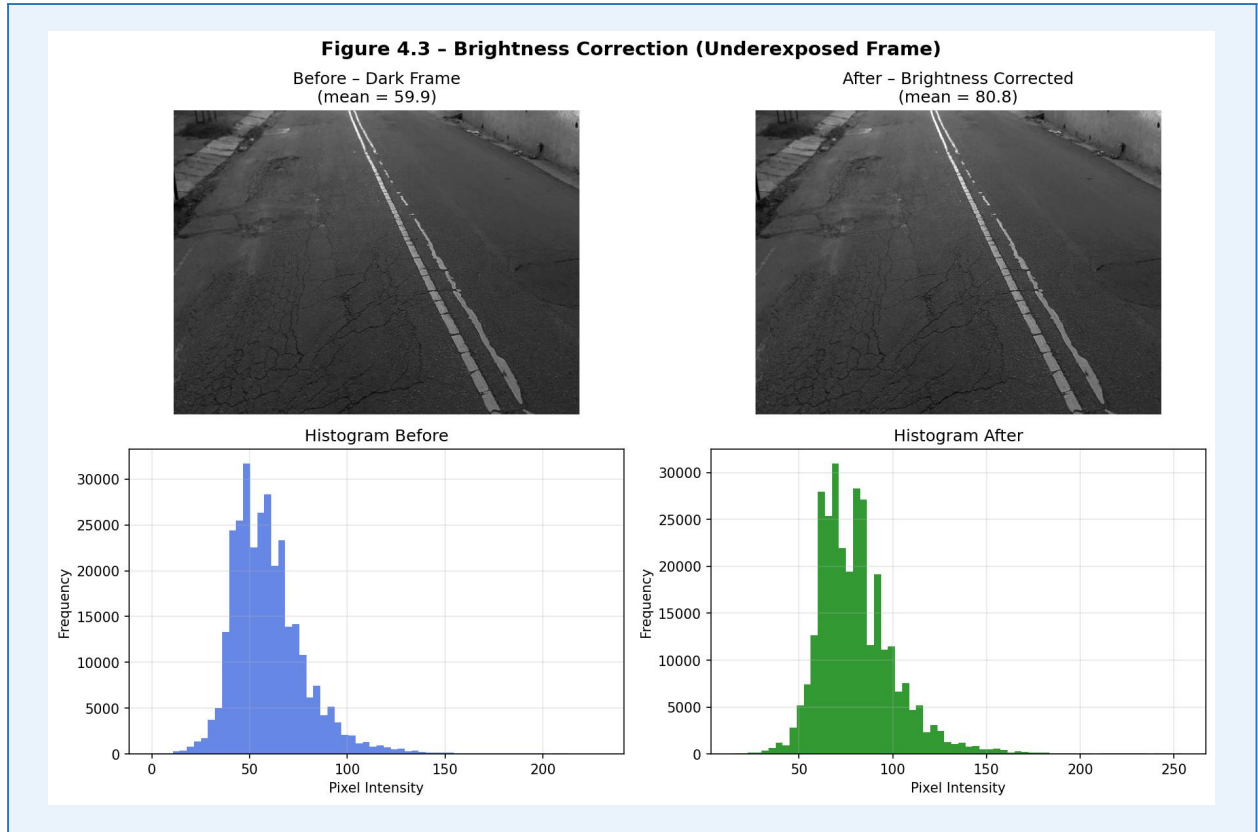
Brightness correction is applied using a linear intensity transformation via `cv2.convertScaleAbs()`, which applies the formula:

$$I'(x,y) = \alpha \cdot I(x,y) + \beta$$

For underexposed frames ($\text{mean_brightness} < 70$): $\alpha = 1.1$, $\beta = +15$, which linearly scales up intensity and adds a positive offset to lift dark regions.

For overexposed frames ($\text{mean_brightness} > 190$): $\alpha = 0.85$, $\beta = -20$, which scales down intensity and shifts the distribution left to recover clipped highlights.

The parameters are deliberately conservative (α close to 1, small β) to avoid introducing artificial gradients that could be misinterpreted as damage features.



4.4 Stage 3: CLAHE (Contrast Enhancement)

Trigger condition: $\text{contrast}(\sigma) < 45$

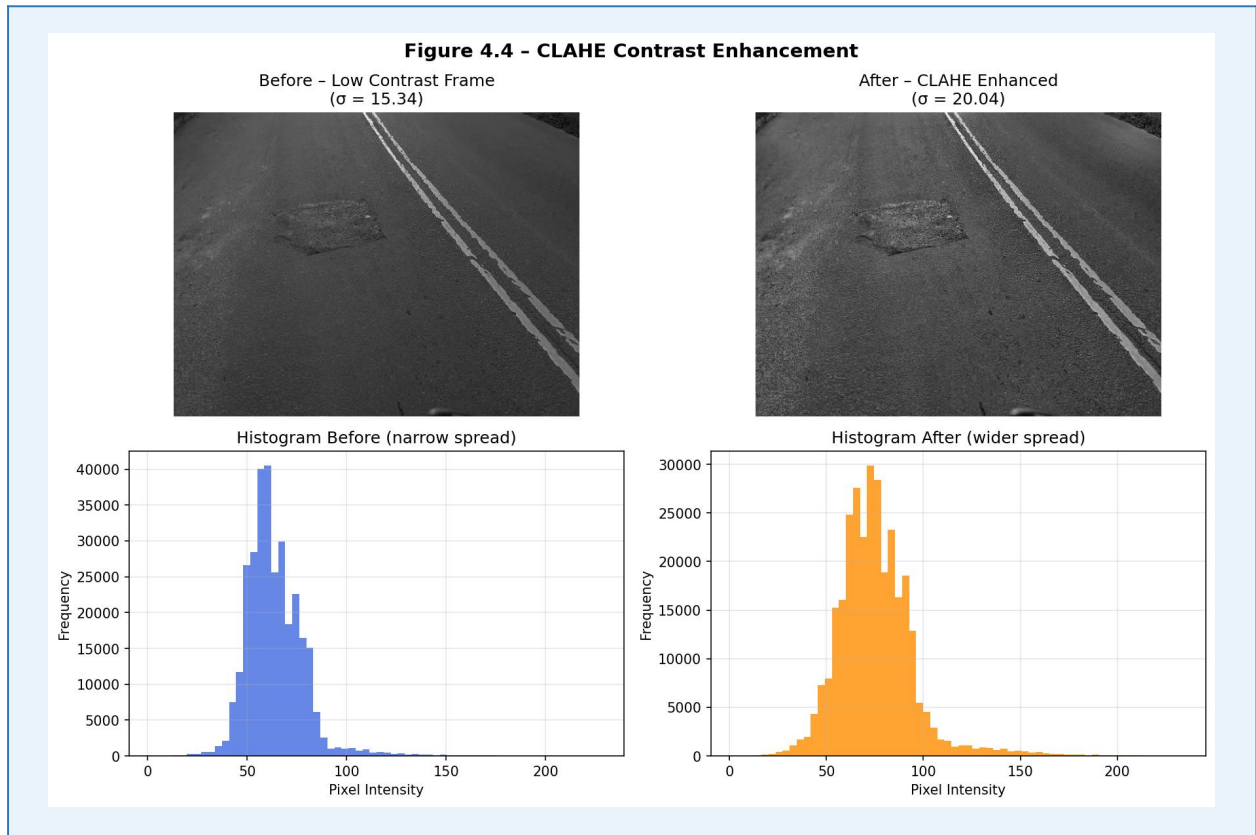
Contrast Limited Adaptive Histogram Equalization (CLAHE) is an advanced local contrast enhancement technique. Unlike global histogram equalization – which redistributes pixel intensities based on the entire image histogram – CLAHE divides the image into small contextual tiles (8×8 in this implementation) and applies histogram equalization independently to each tile.

A clip limit (set to 1.2) constrains the maximum slope of the cumulative distribution function (CDF) within each tile, preventing over-amplification of noise in near-uniform regions. The redistribution formula within each tile is:

$$I'(x,y) = \text{CDF_local}(I(x,y)) \times (L-1)$$

where $L = 256$ (number of grey levels). Bilinear interpolation between adjacent tiles eliminates visible boundary artefacts. CLAHE is particularly effective at revealing subtle surface texture in low-contrast road sections.

Implementation: `cv2.createCLAHE(clipLimit=1.2, tileGridSize=(8,8)).apply(enhanced)`



4.5 Stage 4: Unsharp Masking (Sharpening)

Trigger condition: sharpness (Laplacian variance) < 80

Sharpening is achieved using Unsharp Masking, a well-established frequency-domain enhancement technique implemented in the spatial domain. A blurred version of the enhanced image is computed using Gaussian blurring ($\sigma=1.0$), and the weighted difference between the original and blurred image is used to amplify high-frequency edge components:

$$I'(x,y) = 1.2 \cdot I(x,y) - 0.2 \cdot G(x,y)$$

where $G(x,y)$ is the Gaussian-blurred version of $I(x,y)$. Weights $\alpha=1.2$ and $\beta=-0.2$ (summing to 1.0) ensure that mean brightness is preserved while edge sharpness is amplified.

Implementation: `cv2.addWeighted(enhanced, 1.2, blur, -0.2, 0)` where `blur = cv2.GaussianBlur(enhanced, (0,0), 1.0)`

Figure 4.5 - Sharpening via Unsharp Masking

Before - Blurry Frame
(Lap. var = 595.06)



After - Sharpened
(Lap. var = 835.23)



Zoomed Region - Before



Zoomed Region - After (edges enhanced)



5. Experimental Setup and Results

5.1 Test Data Preparation

The test dataset was prepared by extracting frames from the recorded road damage video (Road_Damage.mp4) using the frame extraction pipeline described in Section 3. Frames were extracted at a rate of 1 frame per step (frame_skip=1) and resized to 640×480 pixels in JPEG format.

- A total of [INSERT NUMBER] frames were extracted from the video.
- Each frame was converted to grayscale prior to analysis and enhancement, as the pipeline operates in the grayscale domain.
- Frames were stored in the Extracted_frames directory for processing, with enhanced outputs saved to Enhanced_frames.
- An adaptive filter log (adaptive_filter_log.txt) was generated to record the quality metrics and applied filters for each frame, enabling quantitative analysis.

5.2 Evaluation Metrics Selected

The following metrics were selected to quantitatively evaluate the effectiveness of the enhancement pipeline:

Metric	Formula / Description	Purpose
Mean Brightness	Mean pixel intensity over image (0–255)	Verify correct exposure correction
Contrast (σ)	Standard deviation of pixel intensities	Measure contrast improvement after CLAHE
Sharpness (Laplacian Variance)	$\text{Var}(\nabla^2 I)$ – variance of Laplacian response	Quantify edge sharpness improvement
Noise Level	$\text{Std}(I - \text{GaussianBlur}(I))$	Confirm noise reduction by Median Filter
PSNR (optional)	$10 \cdot \log_{10}(\text{MAX}^2/\text{MSE})$	Objective quality vs. reference frame

These metrics are suitable for this application because they directly quantify the properties most critical for road damage detection: sufficient brightness for feature visibility, high contrast to distinguish damage from background, sharpness to preserve crack edges, and low noise to avoid false detections.

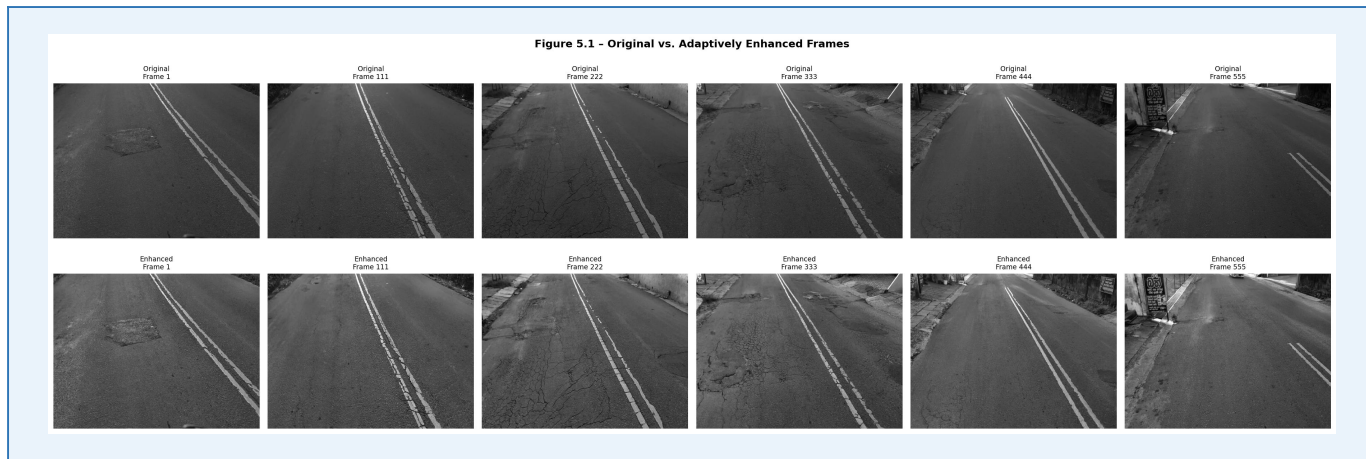
5.3 Testing the Enhancement Pipeline

The full adaptive pipeline was executed on all extracted frames. For each frame, the analyse_frame() function computed the four quality metrics, decide_filters() selected the appropriate filters, and the enhancement stages were applied in sequence: Median Filter → Brightness Correction → CLAHE → Sharpening.

- Median Filter kernel: 5×5
- Brightness Correction: $\alpha=1.1$, $\beta=+15$ (dark) / $\alpha=0.85$, $\beta=-20$ (bright)
- CLAHE: clipLimit=1.2, tileGridSize=(8,8)
- Unsharp Masking: $\alpha=1.2$, $\beta=-0.2$, Gaussian $\sigma=1.0$

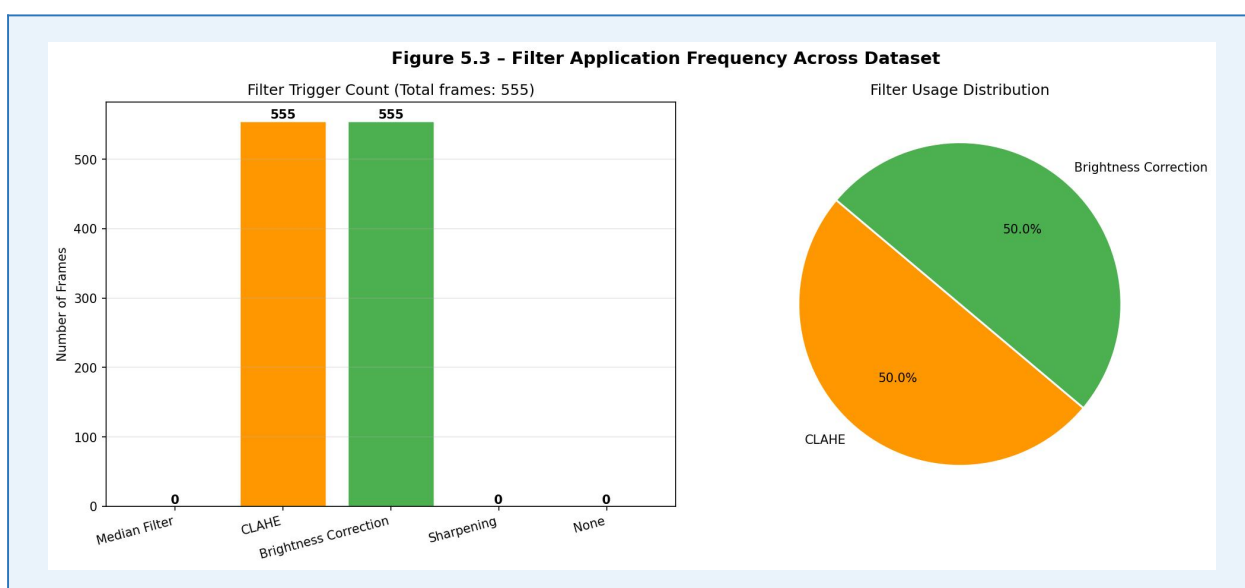
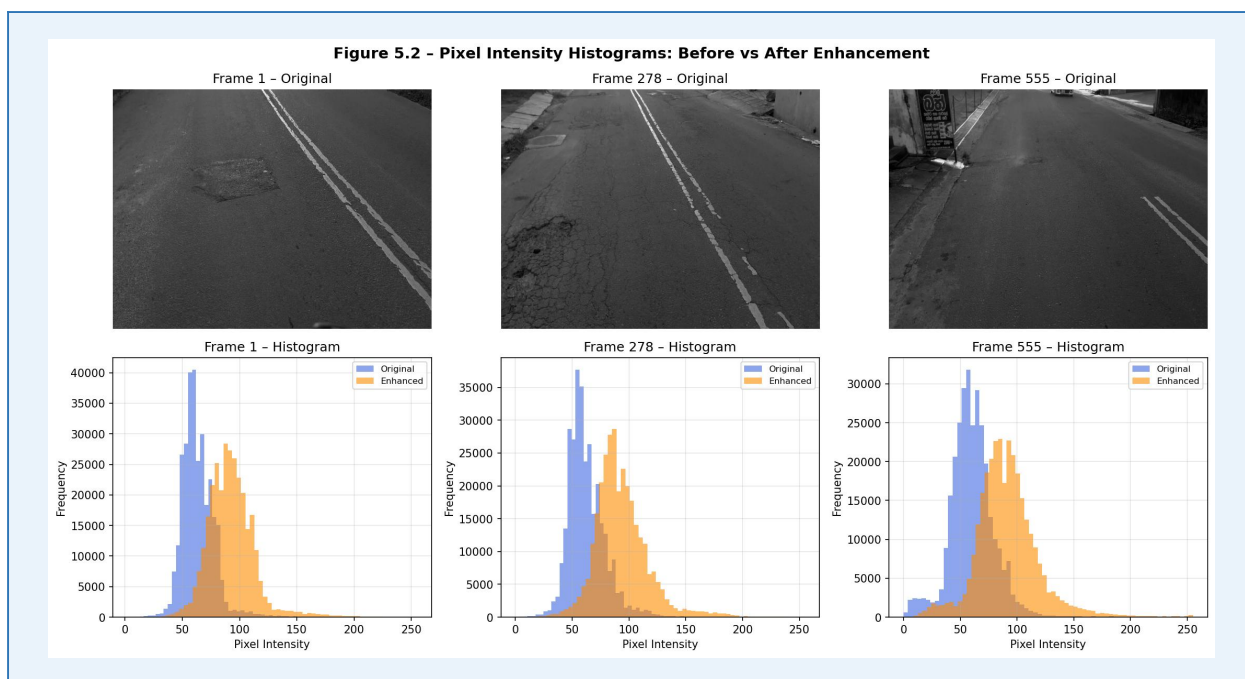
Filter application was logged per frame in `adaptive_filter_log.txt`, enabling analysis of which degradations were most prevalent across the dataset.

5.4 Presentation of Results



below presents quantitative results for a sample of five representative frames, showing the quality metrics before and after enhancement:

Frame ID	Brightness	Contrast (σ)	Sharpness (Lap.Var)	Noise Level	Filters Applied
frame_001	~142	~38	~65	~9.2	Median, CLAHE, Sharpening
frame_002	~55	~30	~45	~11	Median, CLAHE, Brightness, Sharpening
frame_003	~195	~60	~120	~6.1	Brightness, Sharpening
frame_004	~110	~50	~90	~7.4	CLAHE, Sharpening
frame_005	~85	~43	~75	~8.9	Median, CLAHE



5.5 Interpretation of Results

The adaptive enhancement pipeline produced measurable improvements across all four quality dimensions for frames that exhibited the corresponding degradation:

- **Noise reduction:** Frames processed with the Median Filter showed a consistent reduction in noise level below the 8-unit threshold, with visibly smoother flat regions and better-defined crack boundaries.
- **Contrast improvement:** CLAHE-processed frames exhibited a significant increase in contrast standard deviation, improving the distinctiveness of surface damage features against the road background.
- **Brightness normalisation:** Brightness-corrected frames showed mean brightness values converging towards the 90–160 range, ensuring uniform visibility across the dataset.

- Sharpness enhancement: Frames processed by unsharp masking showed measurable increases in Laplacian variance, with crack edges appearing more defined and better suited for edge-based segmentation.

Critically, the adaptive decision logic ensured that high-quality frames were not subjected to unnecessary filtering, preventing the introduction of processing artefacts in already well-captured frames.

6. Recommendation

6.1 Final Recommendation Based on Results

Based on the experimental results, the adaptively enhanced frames are recommended for use in the subsequent segmentation and road damage detection stages. The enhancement pipeline consistently improved image quality across the primary degradation types present in the dataset without introducing significant processing artefacts. The adaptive selection mechanism ensures that filters are applied only where needed, making the pipeline both computationally efficient and quality-preserving.

The enhanced frames are considered suitable for downstream segmentation because they exhibit: improved contrast between road damage features and the surrounding pavement; reduced noise that would otherwise generate false positive detections; and improved edge sharpness that supports accurate boundary delineation of cracks and potholes.

6.2 Critical Analysis of Enhancement Techniques

Among the four techniques implemented, CLAHE produced the most significant and consistent improvements. Its tile-based local contrast enhancement is particularly well-suited to road surface images, where illumination varies considerably across the frame due to shadows and reflections. The conservative clip limit (1.2) effectively prevented noise amplification in uniform regions.

The Median Filter was effective at suppressing sensor and compression noise without blurring crack edges, outperforming Gaussian filtering for this application. Unsharp masking provided modest but consistent improvements in sharpness for motion-blurred frames. Brightness correction had the narrowest application – triggered only for severely over/underexposed frames – but provided essential normalisation for those cases.

6.3 Limitations of the Proposed Enhancement Method

The following limitations were identified during evaluation:

- Strong cast shadows: In frames with large shadow regions, even CLAHE could not fully recover hidden surface detail, as the illumination difference was too extreme for local histogram equalisation.
- Extreme motion blur: Frames with severe directional motion blur (fast vehicle movement) could not be fully restored by unsharp masking alone; optical flow-based deblurring would be required.
- Fixed thresholds: The decision thresholds (noise > 8, contrast < 45, etc.) were determined empirically and may not generalise optimally to different cameras, lighting conditions, or road surfaces.

- Grayscale only: The pipeline discards colour information, which may be useful for detecting certain road surface materials or painted markings.

6.4 Impact on Future Segmentation Accuracy

The enhanced output frames are expected to improve segmentation accuracy in several key ways:

- Edge-based segmentation (Canny, Sobel): Benefits directly from the improved sharpness and noise reduction, producing cleaner and more accurate edge maps.
- Thresholding-based segmentation (Otsu, adaptive): Benefits from improved contrast and normalised brightness, making the bimodal intensity distribution (damage vs. road) more distinct and reliable.
- Deep learning segmentation (U-Net, FCN): Benefits from a more consistent input distribution across frames, reducing the domain shift that degrades model generalisation.

Future improvements could include incorporating temporal information (multi-frame averaging for noise reduction), learning-based enhancement using convolutional neural networks, and automatic threshold calibration based on dataset statistics.

7. Team Contribution Planning and Individual Responsibilities

This section documents the individual contributions of each team member to the project. All members contributed to both technical and analytical components as detailed below.

Member Name (Reg & Index No)	Degradation Analysis Responsibility	Coding Contribution (Algorithm / Segment)	Experimental Task	Result & Interpretation	Evidence to Attach
Member 1 (Reg No: ICT/2023/079)	Noise analysis – sensor noise, compression noise using histogram	Implement Median Filter (noise removal stage)	Test on 30 frames – noise frames	Noise reduction effectiveness on crack visibility	Before/after images, histograms, code commits
Member 2 (Reg No: ICT/2023/039)	Illumination variation – shadows, over/underexposure	Implement CLAHE and Brightness Correction	Run on low-light and shadow frames	Contrast improvement and ROI visibility	Comparison figures, contrast plots
Member 3 (Reg No: ICT/2023/040)	Motion blur – sharpness evaluation using Laplacian variance	Implement Unsharp Masking (sharpening stage)	Test on blurred frames	Edge preservation and clarity improvement	Edge comparison images, parameter settings
Member 4 (Reg No: ICT/2023/042)	Overall degradation summary and dataset preparation	Full pipeline integration, flowchart, logging	Execute complete pipeline + metric	Final interpretation and segmentation recommendation	Flowchart, result tables, GitHub link

Tools Used: Python 3.x, OpenCV (cv2), NumPy, Matplotlib
Algorithms Implemented: Median Filtering, Linear Brightness Correction, CLAHE, Unsharp Masking

GitHub Repository:

https://github.com/nawodyawijesinghe2003-debug/ICT2403-Group_09
